

Isotone Steps of α -Decay Q-Values in the Deformed Actinide Region: A Robust Break at Radium ($Z = 88$)

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Abstract

We report the behavior of the isotone step in alpha-decay Q-values, $\Delta Q(Z,N) = Q_{\alpha}(Z+2,N) - Q_{\alpha}(Z,N)$, across the deformed actinide region ($N > 128$). Using 51 isotone steps derived from 96 Q-values in the AME2020 atomic mass evaluation, we find one robust break, at $Z = 88$ (Ra). Grouped by proton number, the steps have means of 0.308 ± 0.059 MeV ($Z = 84-88$), 0.599 ± 0.179 MeV ($Z = 88-92$) and 1.007 ± 0.302 MeV ($Z \geq 92$), and the pooled groups differ significantly (Welch $t > 5.5$). Pooling, however, mixes neutron numbers. Compared at fixed neutron number, the step across $Z = 88$ rises at every shared N , by 0.090 to 0.465 MeV, which coincides with the onset of permanent octupole deformation near Ra. The apparent second break at $Z = 92$ does not survive the same comparison: at fixed N it changes sign, and it is smaller than the drift of a single boundary's step with N . It reflects the different neutron coverage of the groups, and we do not claim it. $N = 126$ shell-closure effects are excluded by the $N > 128$ restriction.

1. Introduction

The systematic behavior of alpha-decay Q-values across the heavy and superheavy element region has long served as a sensitive probe of nuclear structure [1,2]. Q-values increase monotonically with proton number in the actinide region, reflecting the growing Coulomb instability above the $Z = 82$ shell closure, but the rate and character of that increase varies in ways that encode deformation onset, shell sub-closures, and proton-neutron correlations.

Most systematic studies of actinide α -decay focus on Q-values directly, on half-lives via Geiger-Nuttall relationships [3,4], or on deformation-modified barrier penetration [5,6]. The isotone step $\Delta Q(Z,N) = Q_{\alpha}(Z+2,N) - Q_{\alpha}(Z,N)$ — the change in Q-value when proton number increases by two at fixed neutron number — has received less attention as a primary observable, despite being a natural measure of how rapidly Coulomb instability grows with Z across an isotone chain.

In this paper we compute ΔQ systematically across all available even-even isotone chains in the deformed actinide region ($N > 128$, $Z = 84-100$) using the AME2020 atomic mass evaluation [7]. Grouped by proton number, the 51 available steps fall

into three groups with well-separated means. Compared at fixed neutron number, only the boundary at $Z = 88$ holds, and it coincides with the documented onset of octupole deformation near Ra. The apparent boundary at $Z = 92$ reflects the different neutron coverage of the groups.

2. Data and Method

Dataset. We use atomic mass excesses from the AME2020 evaluation (Wang et al. 2021 [7]) for even-even nuclei from $Z = 82$ to $Z = 102$, $A = 200$ to $A = 260$. The alpha-decay Q-value is computed from:

$$Q_{\alpha}(Z, A) = \Delta(Z, A) - \Delta(Z-2, A-4) - \Delta(^4\text{He})$$

with $\Delta(^4\text{He}) = 2424.92$ keV. This yields 96 individual Q-values across the accessible isotone chains. Seven benchmark Q-values were verified against published experimental values to ≤ 0.2 keV.

The isotone step observable. For each neutron number N and each pair of adjacent even- Z isotones with available Q-values, we compute:

$$\Delta Q(Z, N) = Q_{\alpha}(Z+2, N) - Q_{\alpha}(Z, N)$$

This is the discrete derivative of Q-value with respect to proton number at fixed N , measured in $\Delta Z = 2$ steps.

Deformed region restriction. The $N = 126$ shell closure produces large Q-value steps (~ 0.84 MeV average across all Z boundaries) that dominate and mask the Z -dependent regime structure at $N \leq 128$. We restrict to $N > 128$ (the deformed actinide region) throughout. This yields 51 isotone step measurements (74 over all N). The restriction is essential: without it, shell-closure effects would overwhelm the Z -dependent structure reported here.

Proton coordinate. We parameterize by $U = Z - 80$, measured from mercury ($Z = 80$), so that Po ($Z = 84$) corresponds to $U = 4$, Ra ($Z = 88$) to $U = 8$, U ($Z = 92$) to $U = 12$, and the top of the accessible region (Fm, $Z = 100$) to $U = 20$. The doubly-magic nucleus ^{208}Pb ($Z = 82$) sits at $U = 2$, just below the $Z = 84$ threshold of alpha-emitting actinide precursors.

Statistical analysis. We group steps by the lower- Z boundary: Regime I collects steps with $Z = 84$ – 88 ($U = 4$ – 8), Regime II with $Z = 88$ – 92 ($U = 8$ – 12), and Regime III with $Z = 92$ – 100 ($U = 12$ – 20). For each regime we report the mean $\langle \Delta Q \rangle$, standard deviation σ , and count n . Regime separation is assessed by Welch’s two-sample t -test. Because the groups cover different ranges of N , each boundary is also tested by comparing the steps on either side of it at the same N (Section 3.3).

3. Results

3.1 Grouping by Proton Number

Table 1 summarizes the regime statistics. Figure 1 shows all 51 individual $N > 128$ step measurements plotted against the U coordinate, with regime mean lines and boundaries indicated. The grouping by Z is visually clear; Section 3.3 tests which of its boundaries survive at fixed N .

Table 1. Isotone ΔQ step statistics by regime ($N > 128$ only).

Regime	Z range	U range	$\langle \Delta Q \rangle$ (MeV)	σ (MeV)	n
I	84 $\rightarrow$ 88	4 $\rightarrow$ 8	0.308	0.059	7
II	88 $\rightarrow$ 92	8 $\rightarrow$ 12	0.599	0.179	14
III	92 $\rightarrow$ 102	12 $\rightarrow$ 22	1.007	0.302	30

The ratio of successive regime means is $II/I = 1.94 \approx 2.0$ and $III/II = 1.68$.

3.2 Statistical Significance

The separation between Regimes I and II:

- Difference in means: 0.291 MeV
- Standard error of difference: 0.053 MeV
- $t = 5.51$ (Welch, $df \approx 18$), $p < 0.001$

The separation between Regimes II and III:

- Difference in means: 0.408 MeV
- Standard error of difference: 0.073 MeV
- $t = 5.59$ (Welch, $df \approx 40$), $p < 0.001$

Both pooled differences are statistically significant, so the grouping is not an artifact of small sample size. Pooling, however, mixes neutron numbers; Section 3.3 separates the two.

3.3 Boundaries at Fixed Neutron Number

Within the full dataset ($N = 130$ to 156), the mean ΔQ per isotone step increases with neutron number, with the steepest rise at $N \approx 138$ - 140 (the Ra-226 / Th-230 region). The groups also cover different N ranges: Regime I has $N = 130$ - 136 , Regime II has $N = 130$ - 146 , and Regime III has $N = 134$ - 156 . A difference between pooled group means can therefore reflect N as well as Z . Table 2 compares the steps on either side of each boundary at the same N .

Table 2. Isotone steps on either side of a boundary at equal N (MeV; differences computed before rounding).

N	Rn $\rightarrow$ Ra	Ra $\rightarrow$ Th	Across Z = 88
130	0.343	0.433	+0.090
132	0.331	0.539	+0.207
134	0.273	0.621	+0.348
136	0.199	0.664	+0.465

N	Th $\rightarrow$ U	U $\rightarrow$ Pu	Across Z = 92
134	0.402	0.239	-0.163
138	0.472	0.724	+0.251
140	0.644	0.896	+0.253

N	Th→U	U→Pu	Across Z = 92
142	0.776	1.010	+0.234
144	0.901	1.020	+0.119
146	0.937	0.986	+0.049

Across $Z = 88$ the step rises at every shared N . Across $Z = 92$ the difference changes sign (negative at $N = 134$, positive at $N = 138$ – 146) and falls toward zero at higher N . It is also smaller than the drift of the Th→U step itself, which rises from 0.34 MeV at $N = 136$ to 0.93 MeV at $N = 146$. We therefore claim one robust break, at $Z = 88$. A claim of a break at $Z = 92$ would need a pre-registered test with controlled N coverage.

3.4 Exclusion of Shell-Closure Contamination

At $N \leq 128$ (including the $N = 126$ closed shell), the mean isotone step is ≈ 0.84 MeV regardless of Z , substantially larger than any regime mean and masking the Z -dependent structure. The $N > 128$ restriction cleanly removes this effect. No steps crossing the $N = 126$ boundary are included.

4. Discussion

4.1 $Z = 88$ Boundary: Onset of Permanent Deformation

The transition at $Z = 88$ (Ra) between Regimes I and II coincides with one of the most thoroughly documented structural transitions in the actinide region. The Ra isotopes ($Z = 88$) mark the onset of permanent octupole deformation in even-even nuclei: the octupole shape is theoretically expected at proton numbers near $Z = 88$ and neutron numbers near $N = 134$ [8], and calculations across Ra, Th, U, Pu, Cm, and Cf isotope chains confirm that the most pronounced octupole character occurs around $N \approx 134$, with octupole softness setting in from $N \approx 138$ onward [8]. Quadrupole deformation is also known to set in sharply in the Ra-Th region above the $Z = 82$ shell closure.

It is therefore not surprising that the isotone ΔQ gradient changes character at $Z = 88$: the onset of permanent deformation alters the proton-number dependence of nuclear binding energy, which is exactly what ΔQ measures. The regime change at this boundary reflects a real nuclear structure transition rather than an artifact of the dataset.

4.2 $Z = 92$: Not Supported at Fixed N

The pooled means also change at $Z = 92$ (U). Empirical subdivision of the actinide region at approximately $Z = 89$ – 92 has also been employed for phenomenological Geiger-Nuttall fitting, where a single power-law relationship for the actinides requires separate fitting parameters in the $Z < 92$ and $Z > 92$ regions [9]. In the present data, however, the $Z = 92$ boundary does not survive the fixed- N comparison of Section 3.3: the difference across it changes sign and is smaller than

the drift with N within a single boundary. We therefore do not claim a structural transition at $Z = 92$.

4.3 What the Grouping Reflects

The picture that emerges is physically coherent. Regime I ($Z = 84\text{--}88$) is the region between the $Z = 82$ shell closure and the deformation onset, where nuclei are still relatively close to spherical and the Q -value gradient with Z is modest. Regime II ($Z = 88\text{--}92$) spans the deformation onset region: Ra and Th are in the region of permanent octupole deformation with rapidly evolving structure, producing a more variable and larger mean gradient. Regime III ($Z \geq 92$) covers the well-deformed actinides, where the Q -value gradient with Z is largest and most consistent with systematic Coulomb driving; its higher mean also reflects its wider and higher range of N (Section 3.3).

4.4 The Approximate Doubling Ratio

The $\text{II/I} \approx 2.0$ ratio is notably clean, while $\text{III/II} \approx 1.68$. The III/II ratio compares groups with different N coverage (Section 3.3). The exact ratio values depend on the particular N range accessible in AME2020 for each Z range and may shift as new mass measurements become available, particularly for neutron-rich Ra and Rn isotopes (which contribute to Regime I) where the dataset is thinnest ($n = 7$).

4.5 Limitations

Dataset completeness. Regime I has $n = 7$ measurements — the smallest sample — because isotone chains with both $Z = 84$ (Po) and $Z = 86$ (Rn) data at $N > 128$ are restricted by the short half-lives and measurement challenges for neutron-rich Po and Rn isotopes. The reported Regime I mean should be treated with appropriate caution; new measurements in this region could shift it.

N-correlation. The available isotone chains are not uniformly distributed in N across regimes (Section 3.3). The fixed- N comparison shows that this accounts for the apparent $Z = 92$ boundary, while the $Z = 88$ boundary holds at every shared N . A fully controlled comparison would require equal N coverage across regimes, not yet available from AME2020.

Exclusion of odd- Z isotones. We analyze only even-even nuclei. Odd- Z isotones could in principle show similar structure, but pairing effects and Nilsson-level blocking would complicate the comparison. We make no claim about odd- Z behavior.

5. Conclusions

The isotone step $\Delta Q(Z, N) = Q_\alpha(Z+2, N) - Q_\alpha(Z, N)$ in the deformed actinide region ($N > 128$) shows one robust break, at $Z = 88$, over 51 steps from AME2020. Grouped by Z , the steps form three groups whose pooled means differ significantly ($t > 5.5$), but compared at fixed N only the $Z = 88$ boundary holds: the step across

it rises at every shared N . It coincides with the documented onset of octupole deformation at Ra. The apparent boundary at $Z = 92$ reflects the groups' different N coverage and is not claimed.

This analysis is the first, to our knowledge, to characterize the isotone Q -value step as a primary observable across the deformed actinide series. The $N > 128$ restriction is essential: shell-closure effects at $N = 126$ would mask the structure entirely.

Improved mass measurements for neutron-rich Po and Rn isotopes would sharpen the Regime I statistics, and wider N coverage would allow a controlled test at $Z = 92$.

Note on version 2. Version 1 was deposited on Zenodo on May 29, 2026 (doi:10.5281/zenodo.20448930). This version corrects it: (1) the $N > 128$ step count is 51, not 74 (74 counts all N); (2) only the $Z = 88$ break is claimed, and the $Z = 92$ boundary is shown not to survive a fixed- N comparison (Section 3.3, Table 2); (3) the former Sections 4.5–4.6 are removed: a structural account drawn from a separate framework, and the Curium prediction it motivated, which the data did not confirm. The reference cited for that account did not contain it; (4) an incomplete citation, and the sentence that relied on it, are removed; (5) the abstract's statement that half of the Curium prediction was confirmed is removed with it, since that step showed no significant elevation ($t = +0.4$). The data, the Q -values and Table 1 are unchanged.

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[Figure 1: unchanged from version 1; insert fig1_regime_structure.pdf.]

Figure 1. Isotone Q -value steps $\Delta Q(Z, N)$ in the deformed actinide region ($N > 128$), plotted against $U = Z - 80$ (the lower- Z boundary of each step). Individual measurements are shown as points; horizontal lines are regime means. Shaded regions and colors identify the three groups. Dashed vertical lines mark the group boundaries at $Z = 88$ (Ra) and $Z = 92$ (U); only the first survives the fixed- N comparison of Section 3.3. Element labels at top identify the lower- Z nucleus of each step column.